Tutorial 3 –Experimental error

Background

- When adding and subtracting, the answer should have the same number of decimal places as the limiting term. The limiting term is the number with the least decimal places.
- When multiplying/dividing, the answer should have the same number of significant figures as the limiting term.
- Always retain more digits than necessary during the calculation and round off to the appropriate number of digits in the end.
- Usually, just one significant figure is kept in the error (Never mind it is absolute error or relative error).
- Don't forget that the significant figure of magnitude should be modified by the mantissa of uncertainty for the final result.

Exercises

T3-1. Calculators are great devices. Their invention has allowed for quick computation where manipulation of numbers needs to be done rapidly and accurately. But they are only as good as the numbers put into them. The calculator cannot determine how accurate each set of numbers is, and the answer given must be assessed by the user for reliability.

Consider two separate mass measurements: 0.75 g and 0.241 g, when these masses are added together, the result on a calculator is 0.75 g + 0.241 g = 0.991 g. How should you report the total mass as the answer properly? How many significant figures in your result?

Answer: the total mass is 0.99 g; there are two significant figures.

Explanation: The first mass measurement 0.75 g, is known only to the hundredths place, or to two digits after the decimal point. There is no information about its thousandth place. The second measurement 0.241 g, is known to the thousandths place, or to three digits after the decimal point. Reporting the answer of the total mass as 0.991 g suggests that the sum is known all the way to the thousandths place. However, that cannot be true because the thousandths place of the first mass was completely unknown. The calculated answer needs to be rounded in such a way as to reflect the certainty of each of the measured values that contribute to it. For addition and subtraction calculations, the answer should be rounded to the same number of decimal places as the measurement with the least number of decimal places. The sum of the above masses would be properly rounded to a result of 0.99 g with two significant figures.

T3-2. Check if the significant figures in the final answer are correct. If they are wrong, please correct them.

$$Log\ (1.237 \times 10^{-3}) = -2.908$$

$$10^{2.384} = 242.169$$

$\text{Log}(1.237\times10^{-3}) = -2.908$ False

$= -2.9076$

$10^{2.384} = 242.169$ False

$= 242$

T3-3. a) Calculate the answer to the following equation and give your answer with the correct significant figures.

$$\frac{2.1\times(1.45+0.984)}{8.115\times1.70} - 6.974$$

b) Calculate the absolute and relative uncertainties of the answer to the following equation. Express the answer and the uncertainties with the correct significant figures.

$$6.843\ (\pm\ 0.008) \times 10^4 \div [2.09\ (\pm\ 0.04 - 1.63\ (\pm\ 0.01)]$$

a) $\frac{2.1\times(1.45+0.984)}{8.115\times1.70} - 6.974$

$= \frac{2.1\times2.434}{13.80} - 6.974$

$= 0.37_0 - 6.974$

$= -6.60_4$

≈ -6.60

Always retain more digits than necessary during the calculation (marked by Green color or subscript as shown above) and round off to the appropriate number of digits at the end.

b)

$6.843(\pm\ 0.008) \times10^4 \div [2.09(\pm\ 0.04) - 1.63(\pm\ 0.01)]$

$= 6.843(\pm\ 0.008) \times10^4 \div [0.46_0\ (\pm\ e_3)]$

For $e_3 = \sqrt{e_1^2 + e_2^2} = \sqrt{0.04^2 + 0.01^2} = 0.04123$

So, the equation is,

$= 6.843(\pm 0.008) \times 10^4 \div 0.46_0 (\pm 0.04_{123})$

$= 6.843(\pm 0.117\%) \times 10^4 \div 0.46_0 (\pm 8.963\%)$

$= 148760 \pm \sqrt{0.117\%^2 + 8.963\%^2}$

$= 148760 \pm 8.9638\%$

$= 148760 \pm \frac{148760 \times 8.9638}{100}$

$= 148760 \pm 13334.6$

$= 1.48760 (\pm 0.133346) \times 10^5$

Considering the significant figures, the final result is,

$= 1.5 (\pm 0.1) \times 10^5$ absolute uncertainty

The relative uncertainty is 9%.

$= 1.5 \times 10^5 (\pm 9\%)$ relative uncertainty

T3-4. Write down the answer and uncertainty with the correct significant figures.

a. $\frac{0.002364(\pm 0.000\,003)}{0.02500(\pm 0.000\,05)} =$

b. If the mass is 4.635(±0.02) g and the volume is 8.12(±0.05) mL. How much the density?

c. To prepare a solution, 2.634 ± (0.002) g of NaCl is weighed and dissolved it in a volumetric flask with a volume of 100.00 (± 0.08) mL. The molar mass of NaCl is 58.442(±0.006) g/mol. How much the concentration of NaCl solution in mol/L?

a. 0.09456 (±0.000224)≈0.0946 (±0.0002) or 0.0946±0.2% as the final;

b. the density can be written as 0.5708(±0.00352) g/mL ≈ 0.571(±0.004) g/mL or 0.571g/mL(±0.6%) as the final;

c. The concentration of NaCl is 0.450703(± 0.00049915) mol/L ≈ 0.4507 (± 0.0005) mol/L or 0.4507 mol/L(±0.1%) as the final.

Molarity of NaCl = $\frac{\text{mol of NaCl}}{\text{Volume of solution}}$

$= \frac{\text{mass of NaCl}}{\text{MW (NaCl)} \times \text{Volume of solution}} = \frac{2.634\ (\pm 0.002)\ \text{g}}{58.442(\pm 0.006)\text{g/mol} \times 100.00(\pm 0.08)\ \text{mL}}$

$= \frac{2.634\ (\pm 0.002)\ \text{g}}{58.442(\pm 0.006)\text{g/mol} \times 0.1(\pm 0.00008)\ \text{L}}$

$= \frac{2.634\ (\pm 0.0759\%)\ \text{g}}{8.442(\pm 0.0103\%)\text{g/mol} \times 0.1(\pm 0.08\%)\ \text{L}}$

$= 0.4507_0$ mol/L(± $\%e_4$)

$\%e_4 = \sqrt{\%e_1^2 + \%e_2^2 + \%e_3^2}$

$= \sqrt{(0.0759\%)^2 + (0.0103\%)^2 + (0.08\%)^2}$

=0.1107%

And $e_4 = 0.4507_0$ mol/L × 0.1107% = 0.0004_{99} mol/L

So, Molarity of NaCl = 0.4507 (± 0.0005) mol/L or 0.4507 mol/L(± 0.1%).